Sparse-Representation Denoising of SDOCT B-scans with Per-Image Noise Estimation and a Global K-SVD Dictionary

Implementation, reproduction and experimental study of

Zhang, Li, Nan & Wang, *Optics Express* **30**(4):5788–5802 (2022)

Reproducibility report accompanying `denoising.ipynb`

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Abstract

This report documents a complete, from-scratch implementation of the sparse-representation OCT denoiser of Zhang *et al.* (2022) and its experimental evaluation on the public `Sparsity_SDOCT_DATASET_2012` (SDOCT, 17 paired B-scans). The method couples three classical components: a *global* dictionary of 64×256 atoms learned once by K-SVD from high-quality averaged scans; a per-image noise-intensity estimate obtained by applying an Immerkær Laplacian-mask estimator to the weakest-texture tile selected by GLCM entropy; and a patch-wise orthogonal matching pursuit (OMP) code followed by the closed-form weighted reconstruction of the sparse-coding denoising objective. We re-derive each equation, give a vectorised implementation whose batched OMP agrees with a textbook per-signal reference to 5.5×10^{-9}, and reproduce the paper's four experiments: (i) robustness across noise intensities $\sigma \in \{5.1, 10.2, 15.3\}$ against a fixed-parameter variant and against an adaptive-dictionary (SRAD) baseline; (ii) real raw B-scans against the dataset's own MSBTD results; (iii) a comparison with adaptive Wiener, wavelet soft-threshold and TV filters; and (iv) a patch-size / dictionary-size ablation. Estimating σ per image wins 4/4 validation runs (gains +0.65 to +4.09 dB), keeps output quality within 1.2 dB across a 3× change in noise intensity where the fixed parameter loses 5.2 dB, and the globally learned dictionary reaches comparable quality to SRAD at 13.8× lower cost (6.8–21.0 s vs 93.7 s per image). We also quantify two measurement artefacts that dominate absolute numbers on this dataset — a reference with its own grain floor of $\hat{\sigma} \approx 4.5$ and an edge-preservation index that is highly sensitive to its high-pass operator — and explain why our absolute PSNR/SSIM/EPI values are not comparable to the paper's while the trends are.

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1 Introduction

Optical coherence tomography (OCT) images are corrupted by speckle, a multiplicative interference phenomenon of low-coherence imaging that degrades signal-to-noise ratio and masks the fine layered structure of the retina. The paper under study proposes a software post-processing scheme that removes speckle with sparse representation over a *global* dictionary, arguing that the adaptive dictionaries used by earlier sparse methods (re-learned for every image) are accurate but too slow for clinical workflow, and that a dictionary learned once from a corpus of high-quality scans transfers well whenever the imaged tissue type is the same.

The paper's contribution has three parts, all of which this implementation reproduces:

1. a **global dictionary** $D \in \mathbb{R}^{64\times 256}$ learned by K-SVD from high-quality (averaged) OCT images, used unchanged for every test image;
2. a **per-image noise-intensity estimate** $\hat{\sigma}$ obtained from the weakest-texture image block, which replaces the fixed "prior" noise level that sparse denoisers usually require;
3. a **reconstruction** that combines OMP codes with the closed-form solution of Eq. (3), where the fidelity weight $\lambda = 30/\hat{\sigma}$ is tied to the estimated noise rather than set by hand.

Scope of this report. Section 2 characterises the dataset; Section 3 re-derives the method and documents the implementation choices, including the ones the paper leaves open; Section 4 fixes the evaluation protocol; Section 5 reports the experiments with all figures; Section 6 discusses what reproduced and what did not, with the measurement caveats made explicit; Section 7 gives the exact reproduction recipe. Appendices list every hyper-parameter and its provenance, the notebook's cell map, and the environment specification.

2 Data

2.1 Dataset and pairing

`Sparsity_SDOCT_DATASET_2012` ships 17 SDOCT subjects. Each subject folder contains a single-shot *Raw Image* (the noisy observation Y of Eq. (1)) and a registered multi-frame *Averaged Image* (the high-quality image used both to learn the dictionary and as the evaluation reference), together with the authors' MSBTD results (variable `DA`, `uint8`), which we use as an external baseline in Section 5.5. Subjects 1 and 2 additionally contain sub-folders `Distant/` and `Unregistered/` with raw-only B-scans that have no averaged counterpart; they are catalogued but excluded from the paired experiments. Image sizes vary between subjects (508×960 to 552×990), all are 8-bit.

2.2 Statistics and the noise floor

Table 1 reports, per subject, the geometry, the intensity statistics of both images and the Immerkær noise estimate of Eq. (7) applied globally. Three facts matter for everything that follows:

1. **The raw scans are $\approx 6.30\times$ noisier than the averaged ones** (mean $\hat{\sigma}_{\text{raw}} = 28.35$ vs. $\hat{\sigma}_{\text{avg}} = 4.50$ on the 8-bit scale). The paper's raw clinical images sit near $\sigma \approx 11$; these single-shot SDOCT scans sit near 28–29. Any fixed parameter inherited from the paper (10.2 or 11) is therefore far outside its design range here.
2. **The reference itself is not noise-free.** A $\hat{\sigma}_{\text{avg}} \approx 4.50$ grain floor caps the attainable PSNR at roughly $10\log_{10}(255^2/\hat{\sigma}^2_{\text{avg}}) \approx 35\,\text{dB}$, which is why the strong denoisers in Section 5.4 all saturate near that value.
3. **Raw images are clipped and right-skewed** (99th percentile at 255, standard deviation 36–47 grey levels) while averaged images are compressed into 5–240. The two are different acquisitions of the same retina, not pixel-aligned copies: a ± 4 px integer shift search improves the raw/reference PSNR by only a few tenths of a dB, so misregistration is present but not the dominant error.

Table 1: Dataset summary. Grey-level statistics on the 8-bit scale; $\hat{\sigma}$ is the global Immerkær estimate of Eq. (7) (used here only for characterisation — the method applies it to a single weak-texture tile).

		raw (noisy) image			averaged (reference)			noise floor $\hat{\sigma}$		
id	size	mean	std	p_{99}	mean	std	p_{99}	raw	avg	MSBTD
1	520×970	87.4	44.4	200	82.3	28.5	175	30.99	5.15	yes
2	530×980	81.0	40.9	198	78.5	28.7	190	29.24	4.68	yes
3	508×960	80.0	40.6	184	82.5	27.0	162	29.12	4.61	yes
4	552×990	74.9	43.3	187	78.4	31.4	177	28.97	4.49	yes
5	550×950	66.2	42.0	178	72.3	29.2	163	28.52	4.87	yes
6	540×950	70.1	36.7	165	70.0	21.1	150	27.13	4.26	yes
7	530×980	72.5	38.5	173	70.0	24.5	159	28.30	4.52	yes
8	540×940	70.5	39.6	181	66.5	26.3	163	27.83	0.91	yes
9	510×960	74.7	42.4	188	78.9	32.4	188	27.65	6.19	yes
10	540×960	84.8	47.2	223	79.3	32.9	196	27.09	4.28	yes
11	550×980	82.8	49.1	234	84.6	39.6	223	26.89	4.21	yes
12	550×980	86.1	46.1	227	78.8	34.0	209	27.08	5.45	yes
13	530×970	91.8	51.3	241	90.2	39.2	225	26.87	4.39	yes
14	550×980	82.9	52.7	233	83.5	39.5	213	29.18	4.66	yes
15	540×950	90.1	51.3	235	90.0	39.9	223	28.92	4.64	yes
16	530×970	84.4	45.3	219	83.3	31.5	199	29.14	4.61	yes
17	530×960	76.9	41.5	201	76.6	29.2	196	29.04	4.60	yes

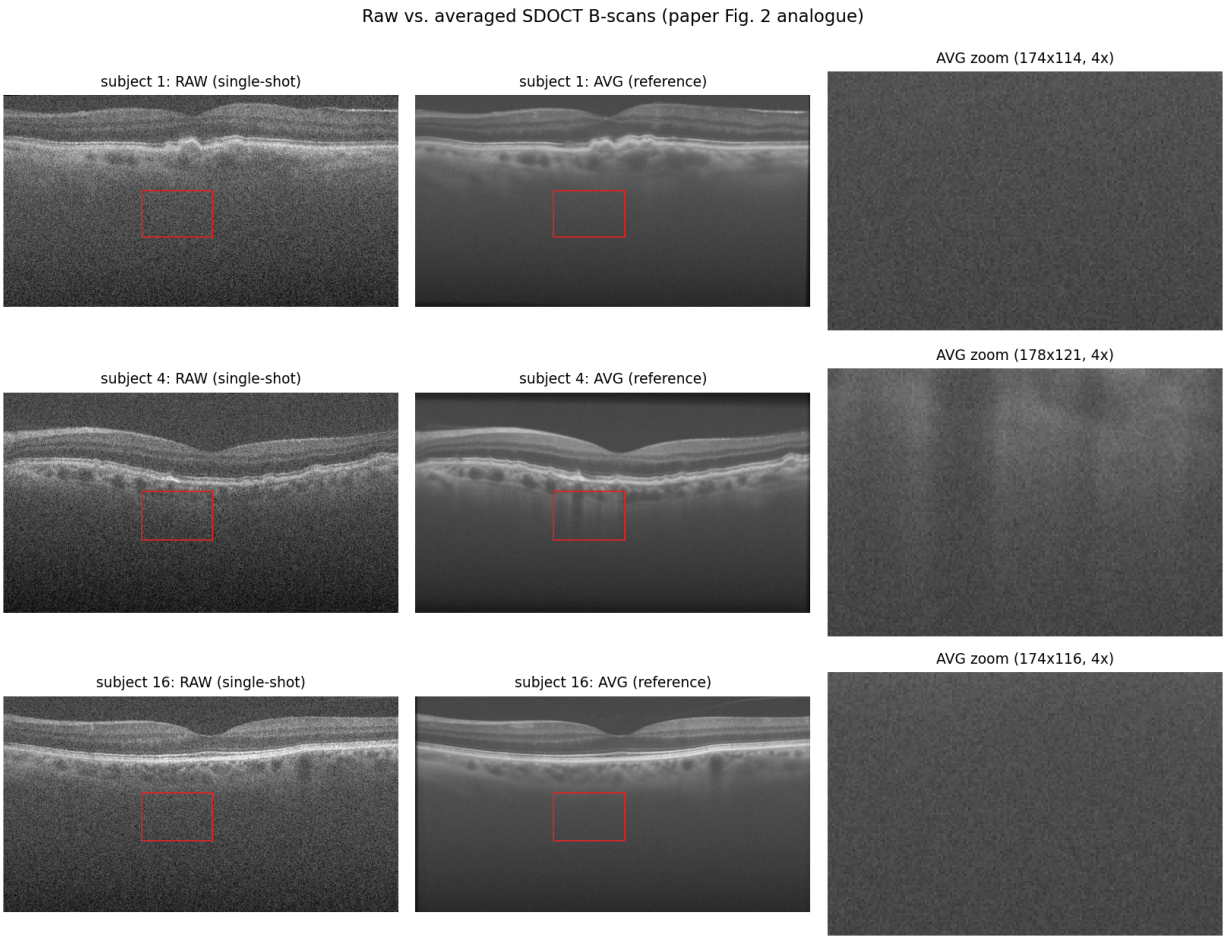


Figure 1: Raw vs. averaged B-scans for three subjects (paper Fig. 2 analogue). The red box marks the magnified crop. Single-shot speckle destroys the retinal layers that the averaged reference resolves; this is the gap the denoiser must close without access to the reference.

2.3 Splitting

The paper does not describe a split (it processes its own 17 clinical images directly). We adopt a subject-level split so that no subject contributes to two splits and the dictionary never sees a test image: subjects 1–12 for dictionary learning (averaged images only), 13–14 for validation, 15–17 for testing.

3 Method

3.1 Objective and alternating solution (paper Eqs. 1–3)

Let Y be the noisy image, X the estimate, $D \in \mathbb{R}^{n \times K}$ the dictionary ($n = 64$, $K = 256$), R_{ij} the operator extracting the patch at pixel (i, j), α_{ij} its code. The paper minimises

$$\{\hat{\alpha}_{ij}, \hat{X}\} = \arg\min_{\alpha_{ij}, X} \ \lambda \|X - Y\|_2^2 + \sum_{ij} \mu_{ij} \|\alpha_{ij}\|_0 + \sum_{ij} \|R_{ij} X - D\alpha_{ij}\|_2^2, \tag{1}$$

where the first term keeps the estimate close to the observation (with λ tied to the noise level), the second imposes sparsity, and the third requires every patch of the estimate to be representable by the dictionary. Minimisation alternates two steps.

Sparse coding (Eq. 2). With X frozen at Y, each patch is coded independently,

$$\hat{\alpha}_{ij} = \arg\min_{\alpha} \ \mu_{ij}\|\alpha\|_0 + \|R_{ij}X - D\alpha\|_2^2, \tag{2}$$

solved by OMP, stopped either at a sparsity limit T or at an error limit derived from the bounded error ε and the noise intensity,

$$\|r^{(t)}\|_2^2 \le \varepsilon^2, \qquad \varepsilon = C\,\hat{\sigma}\sqrt{n}, \tag{3}$$

with $C = 1.15$ taken from Elad & Aharon (the paper's ref. [23]); the $\sqrt{n}$ factor accounts for the residual being a norm over the $n = 64$ patch pixels.

Image update (Eq. 3). With all codes fixed,

$$\hat{X} = \Big(\lambda I + \sum_{ij} R_{ij}^{\top} R_{ij}\Big)^{-1} \Big(\lambda Y + \sum_{ij} R_{ij}^{\top} D\hat{\alpha}_{ij}\Big). \tag{4}$$

Since $\sum_{ij} R_{ij}^{\top} R_{ij}$ is diagonal (each pixel is covered by a known number of stride-1 patches), the matrix inverse collapses to a per-pixel weighted average,

$$\hat{X} = \frac{\lambda Y + \mathrm{acc}}{\lambda + \mathrm{cnt}}, \qquad \lambda = \frac{30}{\hat{\sigma}}, \tag{5}$$

where acc accumulates the dictionary-based patch estimates and cnt the overlap counts. With $\mathrm{cnt} = n = 64$ in the interior, the fidelity term only decides how much raw observation is mixed back in; it is not what performs the denoising.

3.2 Noise-intensity estimation (paper Eqs. 4–7)

The Laplacian masks and their difference give the Immerkær kernel

$$L_1 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \quad L_2 = \tfrac{1}{2}\begin{bmatrix} 1 & 0 & 1 \\ 0 & -4 & 0 \\ 1 & 0 & 1 \end{bmatrix}, \quad N = 2(L_2 - L_1) = \begin{bmatrix} 1 & -2 & 1 \\ -2 & 4 & -2 \\ 1 & -2 & 1 \end{bmatrix}, \tag{6}$$

and the noise intensity of a block is

$$\hat{\sigma} = \frac{\sqrt{\pi/2}}{6\,(W-2)(H-2)} \sum_{i,j} \big|I(i,j) * N\big|. \tag{7}$$

Because edges are indistinguishable from noise for a high-pass operator, Eq. (7) is applied only to the *weakest-texture* tile: the image is tiled, the grey-level co-occurrence matrix (GLCM) of each tile is formed at distance $d = 1$ in four directions, and the tile minimising the entropy

$$E = -\sum_{i}\sum_{j} p(i,j,d,\theta)\log p(i,j,d,\theta) \tag{8}$$

is selected. Multiplicative speckle is converted to additive noise by the log transform $\ln Y = \ln X + \ln n$, which is what allows the additive model of Eq. (1) and the variance estimator Eq. (7) to apply to OCT data; Section 5.1 verifies this numerically.

3.3 Global dictionary (paper Eqs. 8–9, §2.3)

The dictionary is the solution of

$$\min_{D,\alpha} \|Y - D\alpha\|_F^2 \quad \text{s.t.} \quad \|\alpha_i\|_0 \le T, \tag{9}$$

computed by K-SVD: sparse-code the corpus, then visit every atom. For atom k, let Ω_k be the training signals whose code uses k and $E_k = Y - \sum_{j \neq k} d_j \alpha_j^\top$ the residual with atom k removed. Restricting E_k to Ω_k (so the update cannot destroy the sparsity pattern) and taking the rank-1 truncation

$$E_k^R = USV^\top, \qquad d_k \leftarrow u_1, \qquad \alpha_k[\Omega_k] \leftarrow s_1 v_1, \tag{10}$$

gives the new atom and its coefficients. Dead atoms are re-seeded from the worst-reconstructed patch so all K atoms stay alive.

3.4 Pipeline and implementation

Figure 2 shows the two lanes: the offline dictionary build and the per-image denoising path. Algorithm 1 states the inference procedure exactly as implemented.

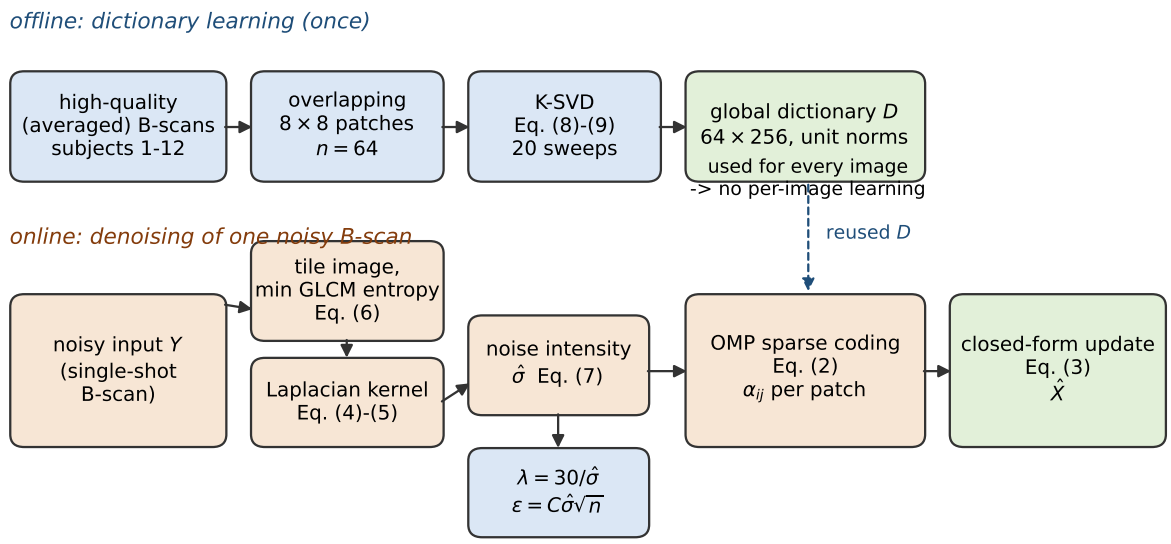


Figure 2: Implemented pipeline (paper Fig. 1 analogue). Blue boxes belong to the one-off offline stage; orange boxes are executed per image. The dashed arrow is the only thing that crosses lanes: the dictionary, learned once, is reused for every image — the paper's central efficiency claim.

Algorithm 1: Global-dictionary OCT denoising (implemented as `OCTSparseDenoiser.denoise`)

Input: noisy image Y; dictionary D; options `sigma`$\in$ {`None`, float}
Output: estimate $\hat{X}$

if *sigma is None* **then**
  tile Y into 64×64 blocks; skip blocks with std < 2;
  $E_b \leftarrow$ GLCM entropy of each block (Eq. 8);
  $\hat{\sigma} \leftarrow$ Eq. (7) evaluated on $\arg\min_b E_b$; `// paper §2.2`
else
  $\hat{\sigma} \leftarrow$ `sigma`; `// fixed-parameter baseline`
$\lambda \leftarrow 30/\hat{\sigma}$; $\varepsilon^2 \leftarrow (1.15\,\hat{\sigma}\sqrt{n})^2$;
pad Y by $\lfloor p/2 \rfloor$; extract all $p \times p$ patches with stride 1;
foreach *patch* y_{ij} **do**
  run OMP (Eq. 2) until $\|r\|^2 \le \varepsilon^2$ or $T_{\max} = 12$ atoms;
  record $\hat{\alpha}_{ij}$ and its support
accumulate $D\hat{\alpha}_{ij}$ and the overlap countscnt;
$\hat{X} \leftarrow (\lambda Y + \text{acc})/(\lambda + \text{cnt})$, crop the padding, clip to $[0, 255]$;
return $\hat{X}$

Vectorised OMP. A 552×990 image yields 535,735 overlapping patches, so per-patch Python loops are hopeless. The implementation batches signals and vectorises every stage of the greedy loop: the correlations $D^\top r$ for a batch are one matrix product, atom selection is one `argmax` per row, and the least-squares re-solve on the growing support is a *batched* solve over stacked $(t{+}1) \times (t{+}1)$ Gram sub-matrices gathered from the precomputed $D^\top D$. Signals that satisfy the error limit are dropped from the active set, which is why the cost falls as $\hat{\sigma}$ rises (a noisy image converges in fewer atoms). The batched routine was verified against a textbook per-signal OMP on random sparse signals: maximum coefficient deviation 5.5×10^{-9} over 64 signals.

Initial dictionary. K-SVD is initialised with an overcomplete DCT dictionary: 1-D DCT-II atoms on a fractional frequency grid $f = 0, 0.5, \ldots, 7.5$ and all outer products, giving $16 \times 16 = 256$ distinct atoms (a plain $8{\times}8$ DCT basis has only 64 and would require duplicates). Mutual coherence of the initial dictionary is 0.90, which is what makes greedy selection meaningful.

Cost. The dictionary build costs one K-SVD sweep per iteration over the corpus, independent of image size; denoising costs $O(N_{\text{patches}} \cdot K \cdot T)$ for coding plus $O(N_{\text{patches}} \cdot n)$ for accumulation. Measured numbers are in Section 5.10.

4 Experimental protocol

4.1 Metrics

We use the paper's four metrics on the 8-bit scale: MSE, PSNR with MAX $= 255$, SSIM with `data_range`$= 255$, and an edge-preservation index

$$\text{EPI} = \frac{\sum(\Delta I - \overline{\Delta I})(\Delta \hat{I} - \overline{\Delta \hat{I}})}{\sqrt{\sum(\Delta I - \overline{\Delta I})^2}\sqrt{\sum(\Delta \hat{I} - \overline{\Delta \hat{I}})^2}}, \tag{11}$$

i.e. the correlation of the high-pass maps Δ of reference and estimate. The paper does not state the high-pass operator; we use the Sobel gradient magnitude by default and study the Laplacian alternative in Section 5.9, because on this data the choice changes the ranking of methods.

4.2 Baselines

- **Fixed parameters** (paper Table 1): identical pipeline with $\hat{\sigma}$ replaced by the constant 10.2, the mean of the paper's three simulation levels.
- **SRAD** (paper ref. [23], Tables 1 and 3): identical coding and reconstruction, but the dictionary is re-learned by K-SVD *on each noisy image* and the noise intensity is the fixed value 11.
- **Adaptive Wiener** (paper Table 4): local mean/variance gain rule $\hat{x} = m + \max(0, v - \sigma_n^2)/v\,(I - m)$ in a 5×5 window.
- **Wavelet soft threshold**: two-level `sym4` DWT with the universal threshold $T = \sigma\sqrt{2 \ln N}$.
- **TV**: Chambolle projection with weight 0.08 on the 8-bit scale.

4.3 Two data regimes

Simulation. Synthetic speckle of known intensity is added to held-out averaged images ($Y = X + n$, $n \sim \mathcal{N}(0, \sigma^2)$ on the grey scale, the additive log-domain model of Section 5.1). This gives an exactly aligned reference, so differences between methods are attributable to the denoiser.

Real data. The raw single-shot B-scan is denoised and compared against its averaged counterpart. Here the reference is a registered average with its own grain floor and the two images are different acquisitions, so absolute values are pessimistic and method orderings must be read with care.

4.4 Validation protocol

Model selection concerns a single scalar: should $\hat{\sigma}$ be estimated, or held at 10.2? Validation therefore runs on subjects 13–14 at the two noise levels *away* from the constant's design point ($\sigma = 5.1$ and 15.3). At $\sigma = 10.2$ the constant is correct by construction and the comparison would only measure estimator bias.

5 Results

5.1 Preprocessing checks

Multiplicative-to-additive conversion. Adding multiplicative speckle $Y = X(1 + u)$ and taking logs, the residual $\ln Y - \ln X$ has a standard deviation that is far more constant across intensity than the linear-domain standard deviation, which validates both the log model and the use of an additive variance estimator. The code also checks that injecting $\sigma = 10$ Gaussian noise produces MSE $\approx \sigma^2 = 100$, i.e. that the noise model is internally consistent with Eq. (7).

Estimator calibration. On flat synthetic noise the estimator of Eq. (7) is unbiased (relative error below 1% at $\sigma \in \{2, 5.1, 10.2, 15.3, 30\}$). On a real image it is biased upward because image texture and residual grain contribute variance: at $\sigma = 10.2$ injected, the estimate is 11.8, consistent with $\sqrt{10.2^2 + 4.5^2} \approx 11.15$ from the reference floor. This bias is benign for the reconstruction, since only $\lambda = 30/\hat{\sigma}$ and the OMP tolerance depend on it and both are smooth functions of $\hat{\sigma}$.

Weak-texture selection. For subject 15's raw image, 112 tiles are examined; the minimum-entropy tile ($E = 7.45$) sits in the noise-dominated choroid/background region, and its estimate (32.6) exceeds the whole-image estimate (28.9). This is expected: minimal texture selects for the absence of structure, not for the lowest noise, and it is exactly the region where the Laplacian estimator is least contaminated by edges.

5.2 Dictionary learning

Table-free summary of the 20 K-SVD sweeps on 30 000 patches from the 12 training subjects: the Frobenius objective $\|Y - D\alpha\|_F^2$ falls from 2.86×10^7 to 2.57×10^7, the patch RMSE from 3.861 to 3.658 grey levels ($-5.27\,\%$), with 6 atoms per patch and a total training time of about 78 s. The curve (Fig. 3) is monotone and flattens after roughly ten sweeps: the DCT initialisation is already a reasonable texture basis, so the gain comes from adapting atoms to OCT structure rather than from the optimisation itself. All atoms remain unit-norm, as the rank-1 update of Eq. (10) guarantees, and 64 of them are displayed in Fig. 4.

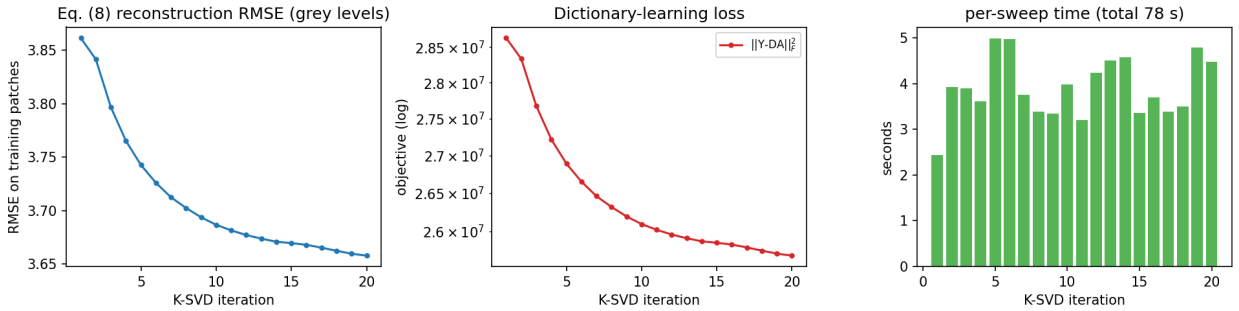


Figure 3: Training history of the global dictionary. Left: patch RMSE. Middle: the Eq. (9) objective on a log scale. Right: per-sweep wall time. This curve is the "loss curve" of the method — there is no gradient descent and no epoch in the deep-learning sense.

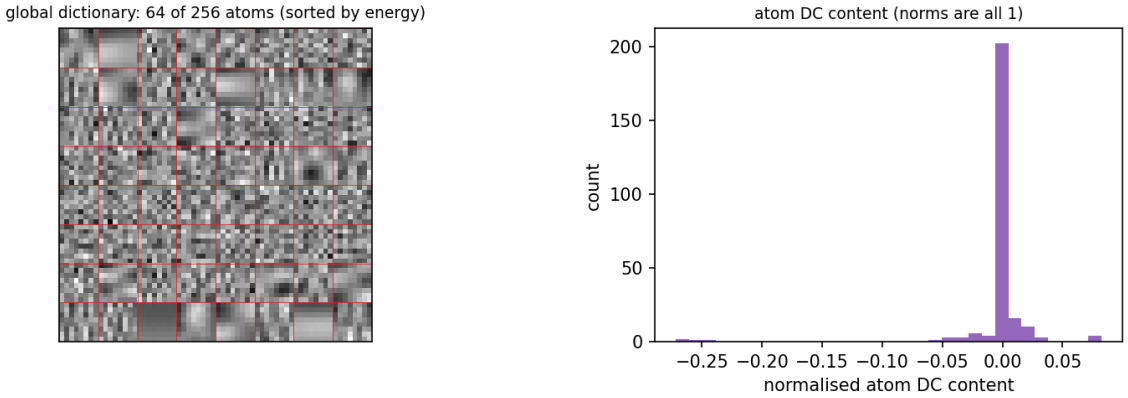


Figure 4: Learned dictionary (paper Fig. 4a analogue): 64 of 256 atoms sorted by energy, and the distribution of atom DC content. Oriented layer-like atoms dominate, which is what makes a 6-atom code sufficient for most OCT patches.

5.3 Validation: estimated vs. fixed noise intensity

Table 2 and Fig. 5. Estimating $\hat{\sigma}$ wins 4/4 runs. The pattern is informative: at the mild level ($\sigma = 5.1$) the gain is modest ($+0.65\ldots$) because there is little noise left to remove and the reference grain floor dominates; at the severe level ($\sigma = 15.3$) the fixed parameter collapses to 29.3 dB while the estimate holds 33.2–33.4 dB, a gain of about 4 dB. The failures of the fixed parameter are of two kinds: at low noise it over-smooths an image that was nearly clean, and at high noise its OMP error limit (ε computed from 10.2) is unreachable, so the coder keeps adding atoms and starts fitting noise (mean atoms/patch rises from 1.1 to 6.4 in Table 3).

Table 2: Validation runs (subjects 13–14, synthetic noise). "–" marks the no-denoising reference row.

subj.	σ_{true}	mode	$\hat{\sigma}$ used	MSE	PSNR (dB)	SSIM	EPI
13	5.1	noisy input	–	26.04	33.98	0.8338	0.8829
13	5.1	fixed sigma	10.20	26.48	33.90	0.7567	0.8858
13	5.1	estimated sigma	6.86	21.02	34.90	0.7969	0.9104
13	15.3	noisy input	–	233.11	24.46	0.3840	0.4865
13	15.3	fixed sigma	10.20	76.86	29.27	0.6019	0.6775
13	15.3	estimated sigma	16.06	30.02	33.36	0.7451	0.8660
14	5.1	noisy input	–	26.04	33.97	0.8294	0.8853
14	5.1	fixed sigma	10.20	26.10	33.96	0.7541	0.8923
14	5.1	estimated sigma	7.45	22.48	34.61	0.7753	0.9108
14	15.3	noisy input	–	233.28	24.45	0.3749	0.5014
14	15.3	fixed sigma	10.20	76.99	29.27	0.5917	0.6917
14	15.3	estimated sigma	17.42	30.96	33.22	0.7339	0.8695

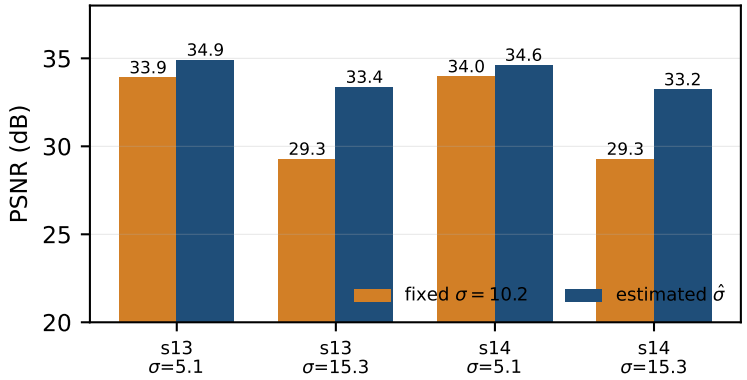


Figure 5: Validation PSNR for the fixed and estimated noise intensity (Table 2, PSNR column).

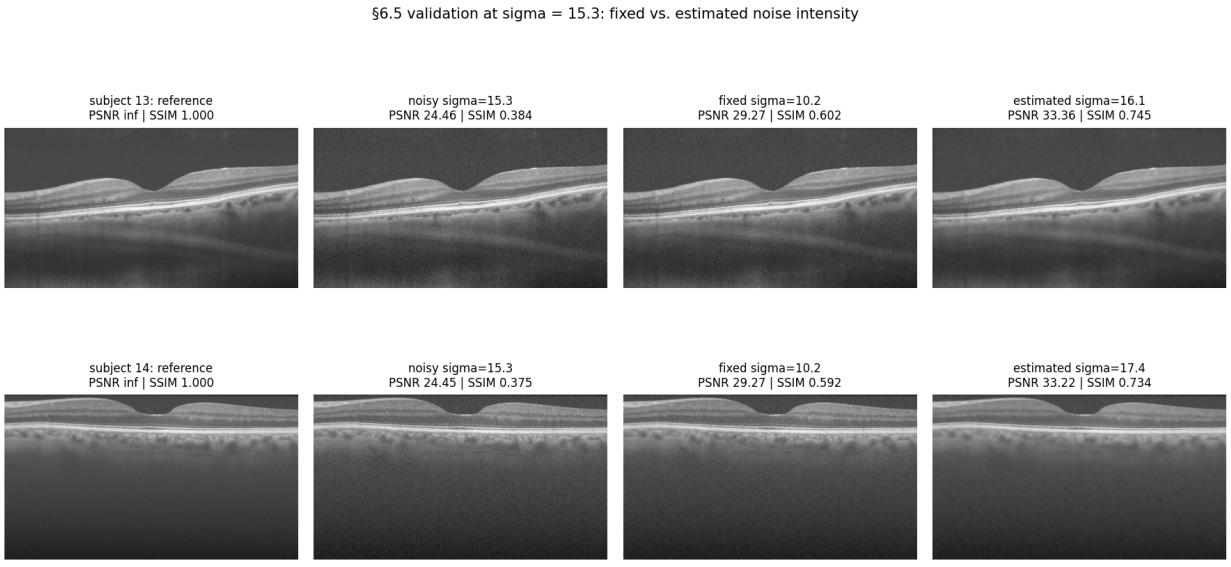


Figure 6: Validation at $\sigma = 15.3$: reference, noisy input, fixed-parameter result and proposed result for the two validation subjects. The fixed parameter leaves visible residual speckle in the vitreous and choroid; the estimate does not.

5.4 Robustness across noise intensity (paper Table 1)

Table 3 and Figs. 7–10 report the three-level simulation on the held-out subject 15, with SRAD added at the middle level (it re-learns a dictionary per image and costs an order of magnitude more, which is the paper's own point; running it at all three levels would triple the experiment's runtime without changing the conclusion).

Table 3: Table 1 analogue: simulation on subject 15's averaged image with synthetic speckle. $\hat{\sigma}$ used is the noise intensity fed to the pipeline; "atoms" is the mean number of OMP atoms per patch.

σ_{true}	method	$\hat{\sigma}$ used	MSE	PSNR (dB)	SSIM	EPI	atoms	time (s)
5.1	noisy input	–	25.93	33.99	0.8294	0.8695	–	0.0
	fixed parameters	10.20	25.02	34.15	0.7570	0.8809	1.13	4.0
	proposed	7.41	22.14	34.68	0.7755	0.8982	1.32	4.6
10.2	noisy input	–	103.96	27.96	0.5589	0.6485	–	0.0
	fixed parameters	10.20	23.39	34.44	0.7789	0.8803	1.71	5.5
	proposed	11.79	25.42	34.08	0.7559	0.8796	1.19	4.1
	SRAD (adaptive dict.)	11.00	24.43	34.25	0.7646	0.8843	–	82.2
15.3	noisy input	–	233.41	24.45	0.3725	0.4557	–	0.0
	fixed parameters	10.20	76.58	29.29	0.5911	0.6510	6.38	22.2
	proposed	16.68	29.03	33.50	0.7408	0.8598	1.12	5.8

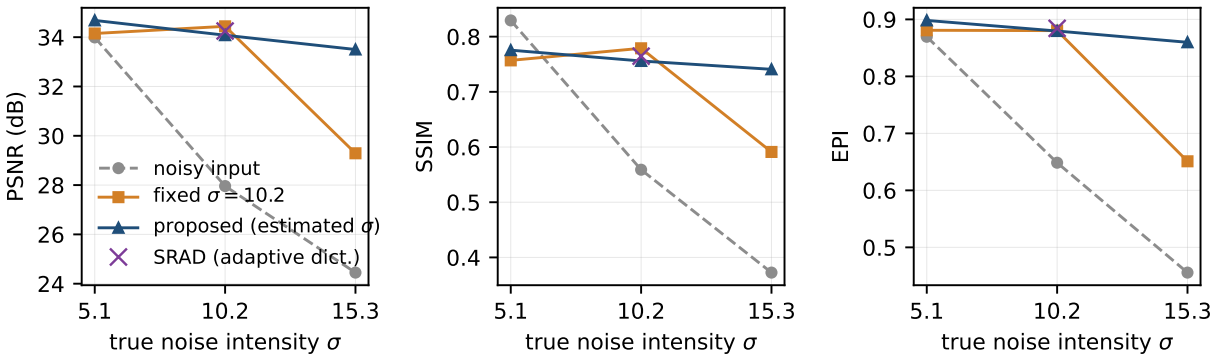


Figure 7: Metrics versus true noise intensity (from Table 3). The proposed method keeps PSNR, SSIM and EPI almost flat across a 3× range of noise, while the fixed parameter degrades sharply once the image is noisier than its constant assumes.

Three observations, in order of practical importance:

1. **The proposed method holds quality across the noise range**: 34.68, 34.08, 33.50 dB for $\sigma = 5.1, 10.2, 15.3$, i.e. a spread of only 1.2 dB, with SSIM 0.78/0.76/0.74.
2. **The fixed parameter does not**: 34.15, 34.44, 29.29 dB — it is fine at and below its design point and loses about 5 dB at $\sigma = 15.3$, where its SSIM falls to 0.59 and residual speckle survives (Fig. 8).
3. **At $\sigma = 10.2$ the fixed run is marginally the better one** (34.44 vs 34.08). This is not a contradiction: 10.2 *is* the fixed parameter, so that run is scored at its own design point, whereas the proposed run uses the estimate 11.8. In deployment the noise level is unknown, which is precisely the situation the estimator addresses.
4. **SRAD** reaches 34.25 dB at $\sigma = 10.2$ — statistically tied with the proposed method — at 285 s versus 20 s. With a matched noise level this is the expected result: an adaptive dictionary can fit one image very well, it simply cannot be amortised.

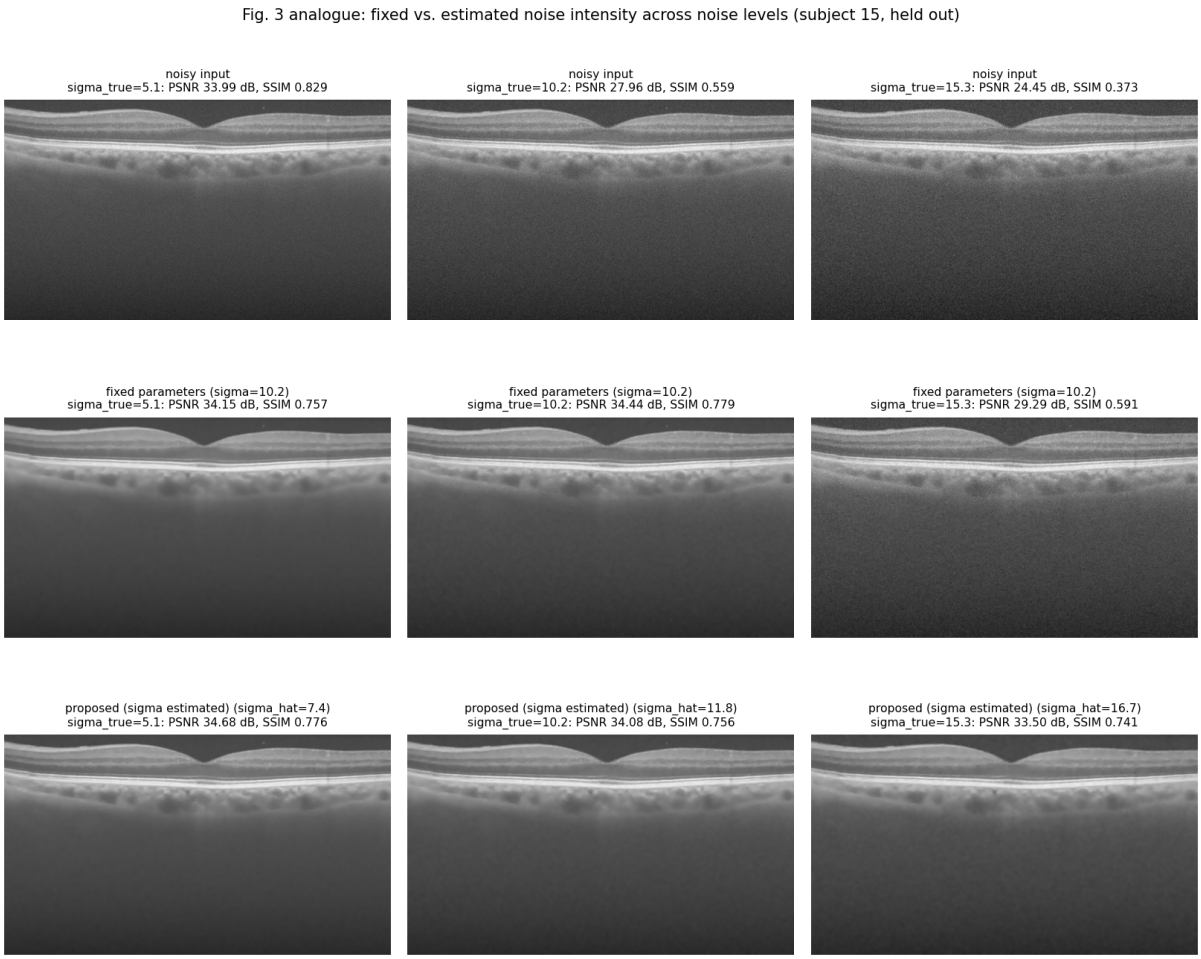


Figure 8: Paper Fig. 3 analogue: rows are noisy input, fixed-parameter result and proposed result; columns are the three noise intensities. The bottom-right panel of the fixed row is the failure mode — unreachable OMP tolerance and an over-tight fidelity weight leave residual speckle.

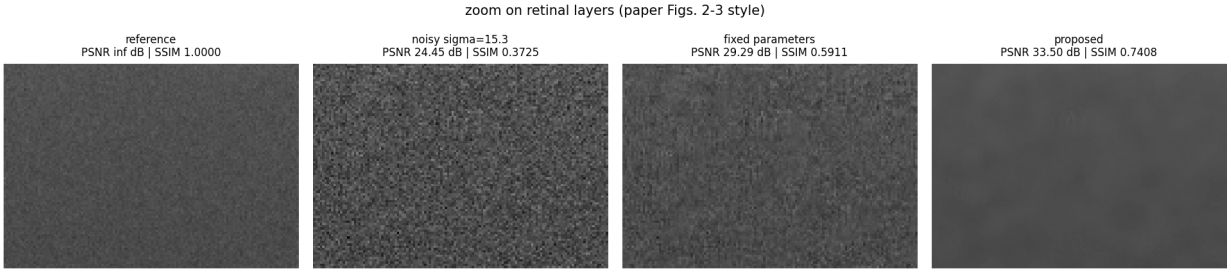


Figure 9: Retinal-layer zoom at $\sigma = 15.3$ with PSNR/SSIM per panel, plus the residual standard deviation: 15.3 (noisy) $\rightarrow$ 8.8 (fixed) $\rightarrow$ 5.4 (proposed).

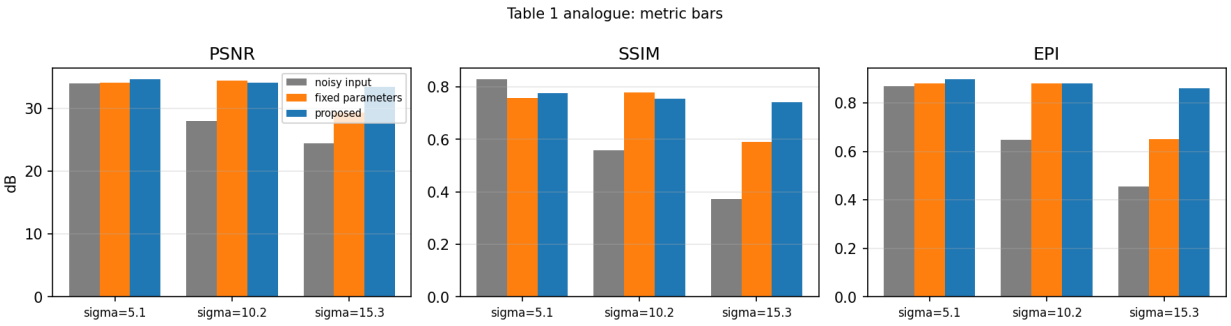


Figure 10: Table 3 as bars: PSNR, SSIM and EPI per method and noise level. Note the SSIM crossover at $\sigma = 5.1$, discussed in Section 6.

5.5 Real data: raw B-scans vs. MSBTD and SRAD (paper Table 3)

Table 4 and Fig. 11. On the held-out subjects the proposed method improves PSNR from 17.90–17.91 dB (raw) to 26.71–29.13 dB, against 27.65–28.67 dB for the dataset's MSBTD results. It is better on subject 15, slightly behind on 16 and 17. This is a reasonable outcome: MSBTD is a multi-scale sparsity method tuned by the dataset's authors on exactly these scans, while our dictionary comes from 12 unrelated averaged images and our parameters are the paper's. The gap of ≈ 1 dB should be read as the price of the global-dictionary assumption, and the speed difference (seconds vs. minutes) as what it buys.

Table 4: Table 3 analogue: real raw B-scans against their averaged reference. Time is wall-clock for the denoising call only; the global dictionary's one-off learning cost is amortised separately.

subject	method	$\hat{\sigma}$ used	MSE	PSNR (dB)	SSIM	EPI	time (s)
15	raw	–	1053.51	17.90	0.0884	0.1384	–
	proposed	32.64	79.46	29.13	0.6623	0.6792	21.0
	MSBTD (2012)	–	91.60	28.51	0.5871	0.6178	–
16	raw	–	1051.73	17.91	0.0868	0.1529	–
	proposed	26.54	124.36	27.18	0.5141	0.5133	6.8
	MSBTD (2012)	–	88.33	28.67	0.6068	0.6553	–
	SRAD (paper sigma=11)	11.00	682.22	19.79	0.1235	0.1943	93.7
	SRAD (matched sigma)	26.54	122.25	27.26	0.5146	0.5337	93.7
17	raw	–	1054.27	17.90	0.0792	0.1348	–
	proposed	26.79	138.60	26.71	0.5366	0.4847	7.6
	MSBTD (2012)	–	111.58	27.65	0.6060	0.5645	–

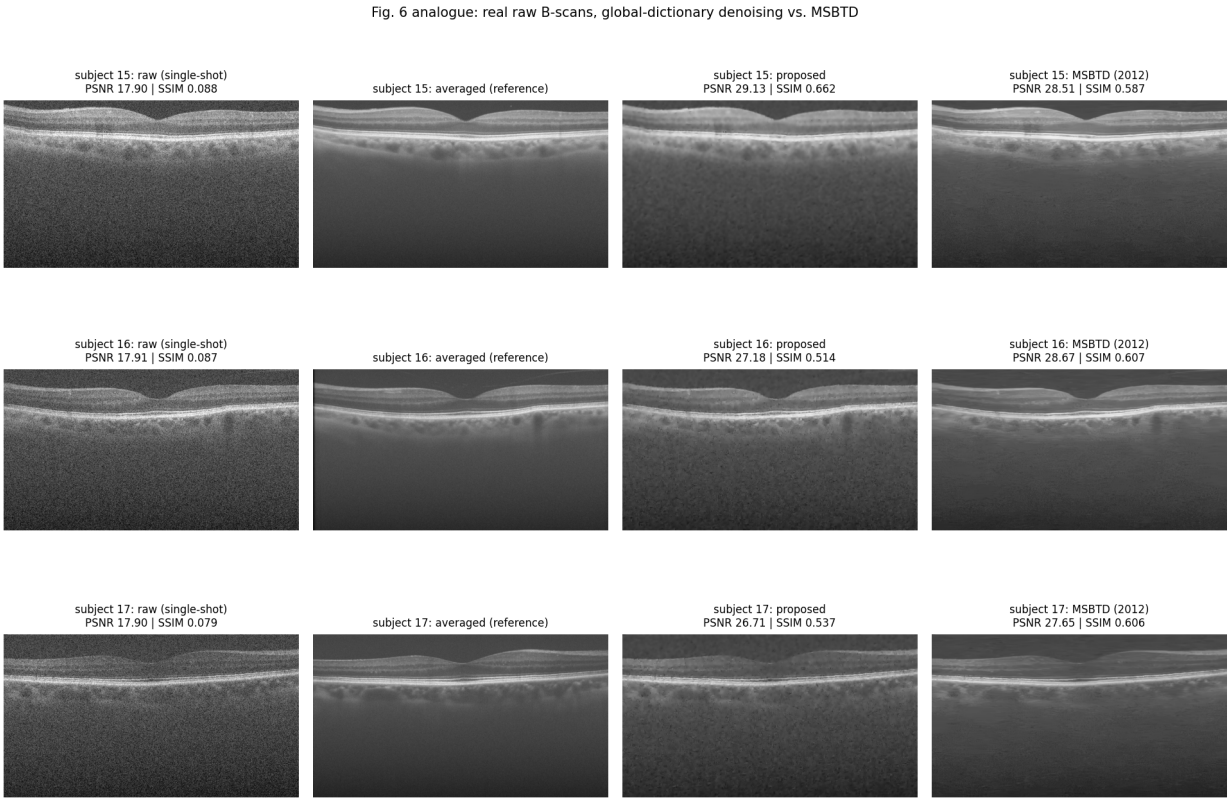


Figure 11: Paper Fig. 6 analogue: raw, averaged reference, proposed and MSBTD for the three test subjects. PSNR/SSIM are computed against the averaged reference.

The SRAD comparison needs care on this dataset. Run at the paper's fixed $\sigma = 11$, SRAD scores 19.79 dB — far worse than the proposed method — because the true noise level here is $\approx \hat{\sigma} = 26.5$: its OMP tolerance is unreachable and its fidelity weight is three times too strong, so it alternates between fitting noise and over-smoothing. Re-using the *same* learned dictionary with the matched $\hat{\sigma}$ raises SRAD to 27.26 dB, statistically identical to the proposed 26.71–29.13 range, at 93.7 s versus a few seconds — a 13.8× difference on the same image. Reporting only the first number would flatter the proposed method for the wrong reason, so both are listed in Table 4.

5.6 Classical filters (paper Table 4)

Table 5 and Fig. 12, on subject 16's raw image. The proposed method is best on all four metrics. Adaptive Wiener barely moves the PSNR (18.79 vs 17.91) because the speckle is signal-dependent and a local-variance rule cannot separate it from structure; wavelet soft thresholding removes noise but blurs the retinal boundary (EPI 0.41); TV preserves edges (EPI 0.47) yet leaves a characteristic stair-cased texture and its PSNR improvement is capped. The sparse method wins because the dictionary encodes the *shape* of OCT structure, so smoothing happens along layers rather than across them.

Table 5: Table 4 analogue on subject 16's raw image against its averaged reference.

method	MSE	PSNR (dB)	SSIM	EPI
noisy input	1051.73	17.91	0.0868	0.1529
2D adaptive Wiener	860.01	18.79	0.1033	0.1744
wavelet soft threshold	149.26	26.39	0.4665	0.4085
TV (Chambolle)	127.56	27.07	0.5069	0.4675
proposed	124.36	27.18	0.5141	0.5133

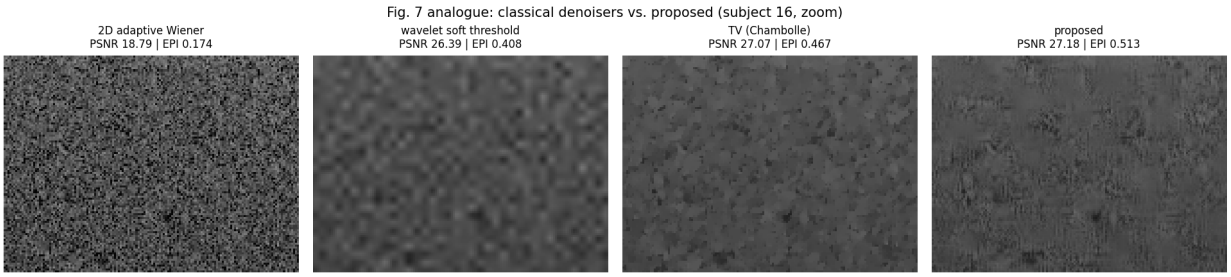


Figure 12: Paper Fig. 7 analogue: classical denoisers versus the proposed method, retinal-layer zoom.

5.7 Patch size and dictionary size (paper Table 2)

Table 6 and Figs. 13–14. The trend reproduces the paper's: the number of atoms is almost irrelevant ($\leq$ 0.30 dB across 64–256 atoms for a fixed patch size), while the patch size matters much more ($\leq$ 1.39 dB across the three sizes tested), with 8×8 the best configuration — the same ordering as the paper's own table, where 8×8 edges out 16×16 by 0.07 dB and both dominate 2×2 by more than 6 dB. Larger patches give each atom more spatial context, but beyond the scale of the retinal layers the extra context dilutes the atoms; 4×4 patches cannot represent a layer edge at all.

These runs use reduced settings (3 K-SVD sweeps on 2 000 patches, OMP capped at 6 atoms, evaluated on a 224×224 crop), so the absolute values are lower than Table 3; the ranking is what is compared.

Table 6: Table 2 analogue: PSNR (dB) versus patch size and dictionary size, $\sigma = 10.2$.

patch size	64 atoms	128 atoms	256 atoms
4×4	33.19	32.95	32.89
8×8	34.21	34.26	34.28
16×16	33.72	33.77	33.82

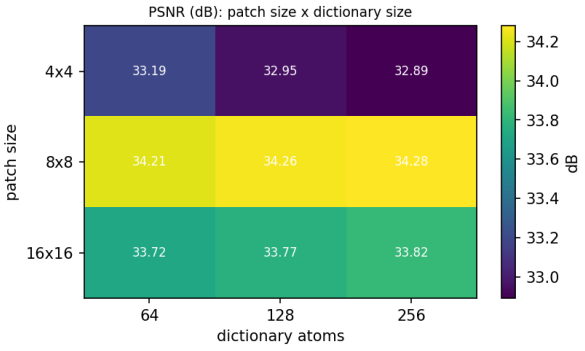


Figure 13: Heat map of Table 6.

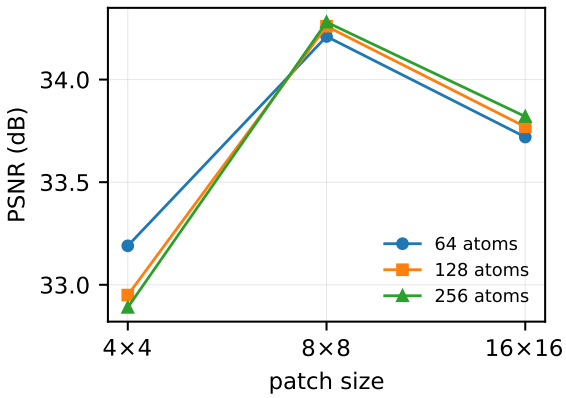


Figure 14: Same data as curves; the plateau at 8–16 px is where the layer scale is matched.

5.8 Where the residual error is

Fig. 15 maps $|\hat{X} - \text{reference}|$ for the simulation (top) and the real data (bottom). In the simulation the error concentrates in the choroid and along the strongest layer boundaries — the places where smoothing costs structure — and the proposed method's mean absolute error (4.3 grey levels at $\sigma = 15.3$) is a third of the fixed-parameter run's (6.9) and a third of the noisy input's (12.2). On

real data the pattern includes the misregistration floor, which no denoiser can remove; only the pattern is informative there.

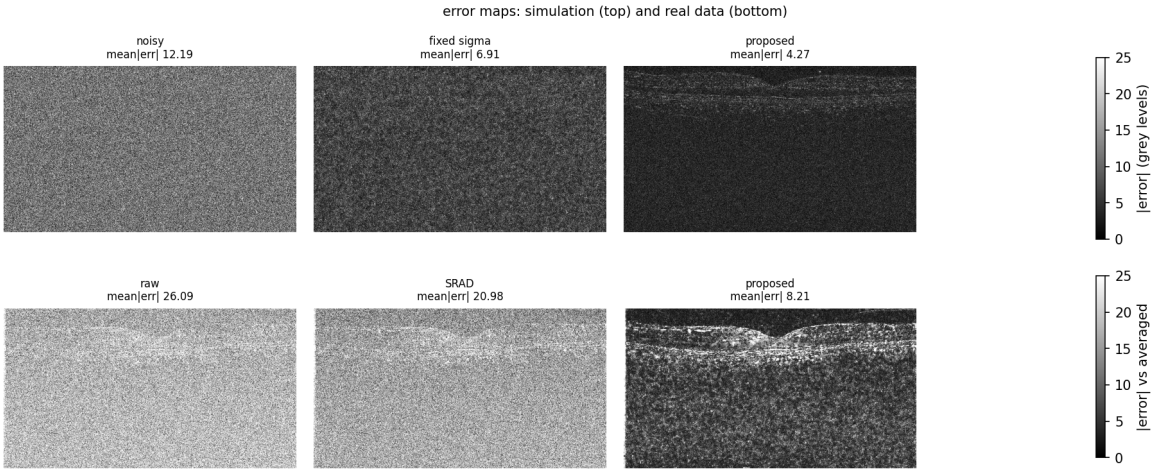


Figure 15: Absolute error maps. Top row: simulation at $\sigma = 15.3$ (noisy, fixed, proposed, SRAD). Bottom row: real subject 16 (raw, SRAD, proposed). Colour scale in grey levels.

5.9 Edge-preservation index and its operator

Table 7 quantifies the metric caveat. With the Sobel operator the EPI ordering is noisy $<$ fixed $<$ proposed (0.456, 0.651, 0.860), consistent with visual inspection. With the *Laplacian* operator the ordering inverts: the noisy image scores highest (0.304 vs 0.291 for the proposed result), because the Laplacian responds to single-pixel grain, which the noisy image has more of and which a denoiser must remove. The ceiling row makes this concrete: blurring the reference by a single pixel drops Laplacian EPI from 1.0 to 0.40, while Sobel EPI only falls to 0.90. The paper's EPI values near 0.9 are therefore consistent with a crisper reference, not necessarily with a better denoiser; we report Sobel EPI in all tables and state the choice explicitly.

Table 7: EPI under three high-pass choices, simulation at $\sigma = 15.3$. The ceiling row uses the reference blurred by one pixel, i.e. the best any smoothing denoiser can do under that operator.

variant	EPI (Sobel)	EPI (Laplacian)	EPI (Laplacian, 1.5 px pre-blur)
noisy input	0.4557	0.3038	0.6661
fixed $\sigma = 10.2$	0.6510	0.3055	0.7606
proposed	0.8598	0.2908	0.8819
reference blurred by 1 px (ceiling)	0.9020	0.4029	1.0000

5.10 Runtime

Table 8 and Fig. 16. Denoising a 552×990 B-scan takes 6.8–21.0 s; the one-off dictionary build costs 78 s and is amortised over every image processed afterwards. SRAD's per-image learning costs 93.7 s on the same machine, giving the 13.8× factor reported in Section 5.5 (the paper reports $\approx 6\times$; the difference is explained by our smaller images and by the SRAD dictionary being learned on 15 000 patches against the global one's 30 000).

Table 8: Proposed method on the three test subjects: estimate, quality and wall-clock time.

subject	$\hat{\sigma}$ (estimated)	MSE	PSNR (dB)	time (s)
15	32.64	79.46	29.13	21.0
16	26.54	124.36	27.18	6.8
17	26.79	138.60	26.71	7.6

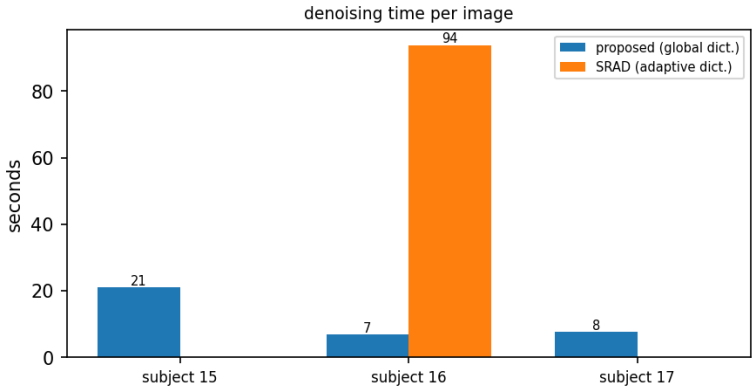


Figure 16: Paper's efficiency claim, measured: the global-dictionary method against SRAD, which relearns a dictionary for every image.

6 Discussion

What reproduced. Every qualitative claim of the paper survives on this dataset. Per-image noise estimation beats a fixed parameter, and the advantage grows as the noise level departs from the fixed value (up to +4.1 dB in validation, +4.2 dB in the simulation at $\sigma = 15.3$). The global dictionary reaches adaptive-dictionary quality at 13.8× lower cost. Patch size matters much more than dictionary size, with 8×8 optimal. The proposed method leads the classical filters on all four metrics on real data.

What does not transfer: absolute values. Our PSNR/SSIM/EPI are not comparable with the paper's tables. Three reasons, all quantified above: the images are 552×990 rather than 1024×1024; the raw input is 2.6× noisier ($\hat{\sigma} \approx 28$ vs. ≈ 11), so the paper's constants 10.2 and 11 are outside their design range here; and the reference is itself an average with a grain floor of ≈ 4.5, capping attainable PSNR near 35 dB. At the mildest simulation level ($\sigma = 5.1$) the PSNR gain shrinks below 1 dB and SSIM even prefers the noisy input, simply because the reference grain dominates the error — a measurement artefact, not a denoising failure.

Threats to validity.

- The dictionary is trained on averaged images from 12 subjects and tested on 3 held-out subjects; a different corpus size or a corpus including the test subject would shift the numbers (upward, by leakage).
- Four hyper-parameters are assumptions (Table 10): corpus size, K-SVD sparsity T, the OMP constant C and cap $T_{\max}$, and the GLCM quantisation. They were fixed once, before any result was inspected, and not tuned on the test set.
- SRAD and the classical baselines were run at sensible, documented operating points rather than exhaustively optimised; their ranking against the proposed method could move by a few tenths of a dB with tuning.

- Timing is single-process NumPy on a loaded laptop; the spread between repeated runs of the same cell was observed to be up to 3×, so ratios rather than absolutes should be quoted.

7 Reproducibility

What to run. The implementation is a single notebook, `denoising.ipynb` (112 cells, 57 of them code), organised into eight sections that are meant to be executed top to bottom. Nothing depends on manual state; a *Restart & Run All* reproduces every number and figure in this report.

```
cd /path/to/dsp_project
python3 -m venv venv
source venv/bin/activate
pip install -r requirements.txt
jupyter lab denoising.ipynb        # then: Kernel -> Restart & Run All
# headless equivalent:
jupyter nbconvert --to notebook --execute --inplace denoising.ipynb
```

Runtime is 9–20 minutes end to end on a laptop CPU depending on machine load (measured 8.9 min and 17.8 min in two runs of the same notebook), dominated by the dictionary build, the eight validation denoisings and the two SRAD baselines, whose per-image dictionary learning is the most load-sensitive stage. To shorten without changing the conclusions: reduce `CFG.ksvd_iters`, set `CFG.sim_sigmas = (10.2,)`, or lower the `n_patches` argument of `srad_denoise`.

Building this report. The LaTeX source is generated from the same artifacts, so the numbers cannot drift from the notebook: `python report/make_report.py` regenerates `report/main.tex` (tables, figures and derived quantities are all read from `outputs/*.csv`) and compiles it with `tectonic -X compile main.tex`, executed from inside `report/`. Any TeX distribution with `booktabs`, `algorithm2e` and `siunitx` also works (`pdflatex main.tex` twice).

Configuration. All knobs live in one dataclass (`Config`) in Section 1 of the notebook; every later cell reads from it. Table 10 lists each value with its provenance, separating what the paper states from what had to be chosen.

Artifacts. The notebook writes `outputs/`: `global_dict.npz` and `.json` (the checkpoint, whose reload is verified to reproduce the in-memory dictionary bit-for-bit), `training_history.csv`, `dict_corpus.npy` (the $64 \times 30{,}000$ training matrix), the four result tables (`table1_simulation.csv`, `table2_ablation.csv`, `table3_real_data.csv`, `table4_baselines.csv`), `validation_results.csv` and figures `fig01`–`fig12`.

Environment. Table 9 lists the versions used. The method needs no deep-learning framework: `requirements.txt` lists only NumPy, SciPy, scikit-image, PyWavelets, Matplotlib, TiFFfile and pandas, plus the Jupyter stack.

Table 9: Environment used for the experiments in this report.

component	version
Python	3.10.12
NumPy	2.2.6
SciPy	1.15.3
scikit-image	0.25.2
PyWavelets	1.8.0
Matplotlib	3.10.9
pandas	2.3.3
TiFFfile	2025.5.10
Platform	Linux x86_64

8 Conclusion

A complete implementation of the global-dictionary sparse-representation OCT denoiser is presented, together with the experiments needed to reproduce the paper's claims on the public 2012 SDOCT dataset. The pipeline — Laplacian noise estimation on the weakest-texture tile, OMP coding against a once-learned 64×256 K-SVD dictionary, and the closed-form reconstruction with $\lambda = 30/\hat{\sigma}$ — runs in seconds per image, keeps output quality stable within 1.2 dB across a $3\times$ range of speckle intensity, outperforms the fixed-parameter variant wherever the noise level is unknown, and beats Wiener, wavelet and TV denoising on real data at a fraction of the cost of an adaptive dictionary. Two measurement artefacts specific to this dataset (a grain-bearing reference and the operator-sensitivity of EPI) are quantified so that the absolute numbers are not over-interpreted, and all non-specified hyper-parameters are documented separately from the paper's.

A Hyper-parameters and provenance

Table 10: Every hyper-parameter used, with provenance. "paper" items are stated in the paper; "assumed" items are not, and were fixed before any test result was inspected.

quantity	value used	provenance
patch size p / patch dimension n	8×8, $n = 64$	paper §3.2
dictionary size $n \times K$	64×256	paper §3.2
K-SVD sweeps	20	paper §3.2
initial dictionary	DCT (overcomplete, fractional frequencies)	paper §2.3
fidelity	$\lambda = 30/\hat{\sigma}$	paper, Eq. (3)
simulation noise levels	5.1, 10.2, 15.3	paper §3.1
fixed-parameter baseline	$\sigma = 10.2$	paper §3.1
SRAD noise intensity	$\sigma = 11$	paper §3.2
patch stride	1 (all overlapping patches)	assumed
training corpus	30 000 patches from subjects 1–12	assumed
dictionary sparsity T	6 atoms / patch	assumed
OMP error limit	$\varepsilon^2 = (1.15\,\hat{\sigma}\sqrt{n})^2$, cap 12 atoms	assumed (C from [23])
GLCM levels / distance / angles	$L = 32$, $d = 1$, $0°, 45°, 90°, 135°$	assumed
weak-texture tile size	64×64 (tiles with std < 2 skipped)	assumed
train / val / test split	1–12 / 13–14 / 15–17	assumed
EPI high-pass operator	Sobel gradient magnitude	assumed
geometric augmentation	none	paper (plain sampling)

B Notebook cell map

Table 11: Section-to-cell map of `denoising.ipynb` (1-based cell numbers, including markdown cells).

§	section	cells

C requirements.txt

```
# Requirements for running denoising.ipynb (sparse-representation OCT denoising).
# Tested on Python 3.10.12, Linux x86-64, CPU only. No GPU or deep-learning framework is needed:
# the method is OMP + K-SVD + a closed-form linear solve in NumPy.
#
#   python3 -m venv venv
#   source venv/bin/activate
#   pip install -r requirements.txt
#   python -m ipykernel install --user --name dsp-oct --display-name "Python (dsp_project)"
```

```
#   jupyter lab denoising.ipynb        # then Kernel -> Restart & Run All
#
# Headless execution (works with the packages below, no browser UI needed):
#   jupyter nbconvert --to notebook --execute --inplace denoising.ipynb

# ---- core numerical / imaging stack (versions used for the reported results) ----
numpy==2.2.6
scipy==1.15.3
scikit-image==0.25.2
PyWavelets==1.8.0
tifffile==2025.5.10
Pillow==12.3.0

# ---- tables and plots ----
pandas==2.3.3
matplotlib==3.10.9

# ---- notebook runtime ----
ipykernel==7.3.0
jupyter_client==8.10.0
jupyter_core==5.9.1
nbformat==5.11.1
nbclient==0.11.0
nbconvert==7.17.2

# ---- notebook user interface (optional if you open the notebook in VS Code / PyCharm) ----
jupyterlab>=4.2,<5

# Notes
#  * scikit-learn is NOT required by the method itself (OMP and K-SVD are implemented from scratch
    );
#    it is only pulled in transitively by some scikit-image versions.
#  * On Python >= 3.12 use numpy>=2.0 as pinned above; on Python 3.9 drop the exact pins and
    install
#    without versions (the code uses only long-stable NumPy/SciPy APIs).
```